

Week 3 Marathon 3 Question 2

Create an enum class with the following literals

- UTILITY, SPORTS, PASSENGER or COMMUTE

Create a class **Car** with the following attributes

- **carPrice** of type float.
- **carBrand** of type string
- **carVehicleType** which is a literal of type VehicleType
- **carSeatCount** which is an integer value between 2 to 7 only

Create a class Bike with the following attributes

- **bikePrice** of type float.
- **bikeBrand** of type string.
- **bikeVehicleType** which is literal of type VehicleType
- **bikeSeatCount** which is either 1 or 2

Create all necessary special member functions (constructors, getter-setters, etc)

Create the following functionalities and demonstrate each by creating client code (main function)

- A function to create a container of instances where each instance could be either a shared pointer to Car or shared pointer to Bike.
- A function to find and return the brand of the Car or Bike instance whose "Price" attribute is the maximum from a container of instances.
- A function to find and return the average Price of all instances passed in a container as an argument.
- A function to find and return a container of **vehicleType** of all **Car** or **Bike** instances (passed in a container) whose price is below a threshold value passed as an argument.
- A function to check whether all **Car** or **Bike** instances passed in a container are of the same **VehicleType** and return a boolean.

- A template function that can accept any container with a **begin** and **end** iterator (vector, array, list, etc) with **Car** or **Bike** instances and prints the attributes of all instances whose Brand matches the second parameter passed to the function.

Note:

- Use C++ 17 features like `std::variant`, `std::optional`, fold expressions, etc wherever applicable.
- All instances are to be created on the heap and their addresses must be stored in Smart Pointers.
- Throw exceptions wherever applicable.